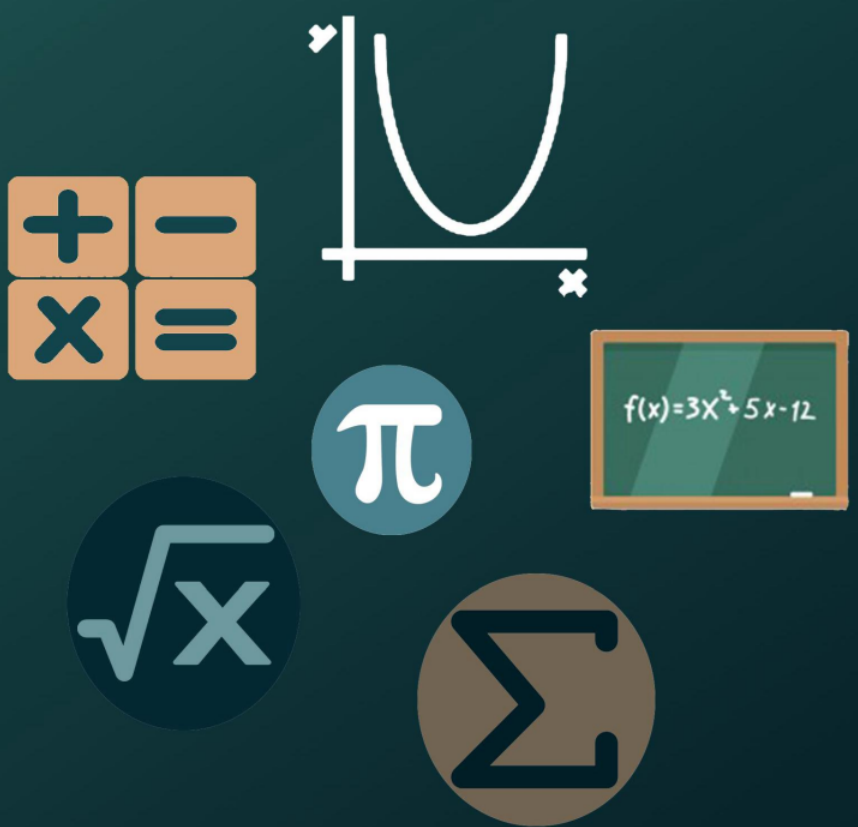
MATHEMATICS

Function



SHEET

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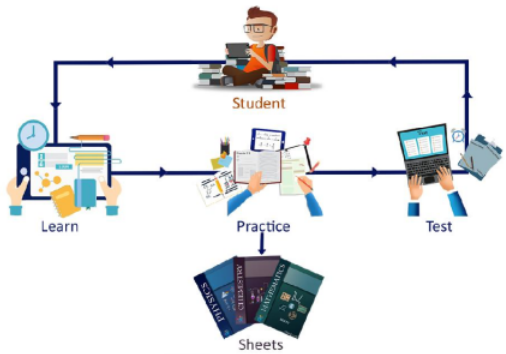
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topics	EX - 1	EX - 1A	EX - 2	EX - 2A
1	Definition	1 - 3	1 - 2	1 - 2	1
2	Value of the Function	4,5			32
3	Domain, Co-domain & Range of the Function	6-11	3, 23-25	3 - 7	22, 33
4	Constant & Identity Function				
5	Signum Function				
6	Exponential & Logarithmic Function				
7	Absolute Value Function				
8	Greatest Integer Function	12	26-30		
9	Fractional Part Function	13			23
10	Polynomial Function				24-26
11	Algebraic & Rational Function	14,15			
12	Algebraic Operations on Function				
13	Range of a Function	16-20	31, 32	8 - 11	19
14	Applications of Functional Rule	21		12 - 16	34, 35
15	Equal or Identical Function	22	4, 33		2
16	Homogeneous Function	23			3
17	Bounded, Implicit & Explicit Function	24			
18	Classification of Function	25-27	16, 34-36	17 - 20	4, 5, 20
19	Composite Functions	28-30	5, 37, 38	21 - 22	
20	Inverse of a Function	31-33	6, 10-13, 39-41		6, 11-13, 27, 28, 36
21	Odd & Even Functions	34	7, 14-15, 42, 43	23	7, 21, 29
22	Periodic Function	35-37	8, 17	24 - 27	8, 9, 30, 37
23	Transformation of the Graphs	38-39	18		10, 38, 39
24	After complete chapter		9, 19-22, 44	28 - 29	14-18, 31

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Definition

Q1. If A = {a, b} and B = {0, 1, 2} then number of functions defined from A to B is-

(A) 9 (B) 8

(C) 6 (D) none of these

Q2. Which of the following relations is a function ?

(A) $x^2 + y^2 = r^2$ (B) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = r^2$

(C) $y^2 = 4ax$ (D) $x^2 = 4ay$

Q3. If A contains 10 elements then total number of functions defined from A to A is

(A) 10 (B) 2^{10}

(C) 10^{10} (D) $2^{10} - 1$

Value of the Function

Q4. If $f(x) = \frac{4^x}{4^x + 2}$, then $f(x) + f(1 - x)$ is equal to

(A) 0 (B) –1

(C) 1 (D) 4

Q5. If $f(x) = \ell n\frac{1+x}{1-x}$ then $f\left(\frac{a+b}{1+ab}\right) =$

(A) f(a) – f(b) (B) f(a) + f(b)

(C) $\frac{f(a)}{f(b)}$ (D) f(a) f(b)

Domain, Co-domain &Range of the Function

Q6. $\sqrt{\sin^{-1}(\log_2 x)}$ exists for

(A) $x \in (1, 2)$ (B) $x \in [1, 2]$

(C) $x \in [2, \infty)$ (D) $x \in (0, \infty)$

Q7. The domain of the function $f(x) = \log_2 \log_3 \log_4 x$ is -

(A) $[4, \infty)$ (B) $(4, \infty)$

(C) $(-\infty, 4)$ (D) None of these

Q8. Domain of the function $y = \frac{\sqrt{3x-7}}{\sqrt[6]{x+1}-2}$ is

(A) [7/3, 63] ∪ (63, +∞)

(B) [7/3, 63) ∪ (63, +∞)

(C) (7/3, 63) ∪ (63, +∞)

(D) None

Q9. Domain of the function $y = \sqrt[4]{x - |x|} + \log(x + 2)$ is

(A) R (B) (0, + ∞)

(C) [0, +∞) (D) None

Q10. Domain of definition of the function f(x) $= \frac{3}{4-x^2} + \log_{10}(x^3 - x)$, is-

(A) (–1, 0) ∪ (1, 2) ∪ (2, ∞)

(B) (1, 2)

(C) (–1, 0) ∪ (1, 2)

(D) (1, 2) ∪ (2, ∞)

Q11. The domain of the function $f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is

(A) [1, 2) (B) [2, 3)

(C) [1, 2] (D) [2, 3]]

Greatest Integer Function

Q12. If $f(x) = \cos\left[\frac{\pi^2}{2}\right]x + \sin\left[\frac{\pi^2}{2}\right]x$, where [·] denotes gretest integer function then :-

(A) $f(0) = 0$ (B) $f\left(\frac{\pi}{3}\right) = \frac{1}{4}$

(C) $f\left(\frac{\pi}{2}\right) = 1$ (D) $f(\pi) = 0$

Fractional Part Function

Q13. If $f(x) = \{x\} + \{x+1\} + \{x+2\}.....\{x+99\}$, then the value of $[f(\sqrt{2})]$ is, where {.} denotes fractional part function & [.] denotes the greatest integer function -

(A) 5050 (B) 4950

(C) 41 (D) 14

Algebraic & Rational Function

Q14. If $f(x) = \frac{1}{\sqrt{x+2\sqrt{2x-4}}} + \frac{1}{\sqrt{x-2\sqrt{2x-4}}}$ for $x > 2$, then $f(11) =$

(A) 7/6 (B) 5/6

(C) 6/7 (D) 5/7

Q15. Domain of the function

$f(x) = \sqrt{\frac{(1-x)(x^2+x+1)}{(x-3)}}$ is :-

(A) $(-\infty, 1] \cup (3, \infty)$ (B) [1, 3]

(C) $(-\infty, 3)$ (D) [1, 3)

Range of aFunction

Q16. The range of the function $f(x) = \frac{x}{|x|}$ is -

(A) R–{0} (B) R–{–1, 1}

(C) {–1, 1} (D) None of these

Q17. The range of the function

$f(x) = \sqrt{3x^2 - 4x + 5}$ is -

(A) $\left(-\infty, \sqrt{\frac{11}{3}}\right]$ (B) $\left(-\infty, \sqrt{\frac{11}{3}}\right)$

(C) $\left[\sqrt{\frac{11}{3}}, \infty\right)$ (D) $\left(\sqrt{\frac{11}{3}}, \infty\right)$

Q18. Range of the function

$f(x) = \log_2\left(\frac{\sin x - \cos x + 3\sqrt{2}}{\sqrt{2}}\right)$ is -

(A) (1, 2) (B) R

(C) [–1,2] (D) [1, 2]

Q19. The range of the function $f(x) = \frac{2+x}{2-x}$, $x \neq 2$ is-

(A) R (B) R – {–1}

(C) R – {1} (D) R – {2}

Q20. The range of the function $f(x) = {}^{7-x}P_{x-3}$ is-

(A) {1, 2, 3, 4, 5} (B) {1, 2, 3, 4, 5, 6}

(C) {1, 2, 3} (D) {1, 2, 3, 4}

Applications of Functional Rule

Q21. Let $f : R - \{0\} \to R$ be any function such that $f(x) + 2f(1/x) = 3x$. The sum of the values of x for which $f(x) = 1$.

(A) 1 (B) 2

(C) –1 (D) –2

Equal & Identical Function

Q22. The functions $f(x) = \log(x-1) - \log(x-2)$ and $g(x) = \log\left(\frac{x-1}{x-2}\right)$ are identical when x lies in the interval

(A) [1, 2] (B) [2, ∞)

(C) (2, ∞) (D) $(-\infty, \infty)$

Homogeneous Function

Q23. Which of the following function is not homogeneous ?

(A) $f(x, y) = x^3 + 8x^2y + 7y^3$

(B) $f(x, y) = y^2 + x^2 + 5xy$

(C) $f(x, y) = \dfrac{xy}{x^2 + y^2}$

(D) $f(x, y) = \dfrac{2x - y + 1}{2y - x + 1}$

Bounded, Implicit & Explicit Function

Q24. Which of the following function is implicit function ?

(A) $y = \dfrac{x^2 + e^x + 5}{\sqrt{1 - \cos^{-1} x}}$

(B) $y = x^2$

(C) $xy - \sin(x + y) = 0$

(D) $y = \dfrac{x^2 \log x}{\sin x}$

Classification of Function

Q25. If $f : R \to R$, $f(x) = x^3 + x$, then f is -

(A) one-one onto (B) one-one into

(C) many-one onto (D) many-one into

Q26. If the function $f : R \to A$ given by $f(x) = \dfrac{x^2}{x^2 + 1}$ is a surjection, then A =

(A) R (B) [0, 1]

(C) (0, 1] (D) [0, 1)

Q27. A function f from the set of natural numbers to integers defined by

$$f(n) = \begin{cases} \dfrac{n-1}{2}, & \text{when n is odd} \\ -\dfrac{n}{2}, & \text{when n is even} \end{cases}$$ is-

(A) neither one-one nor onto

(B) one-one but not onto

(C) onto but not one-one

(D) one-one and onto both

Composite Function

Q28. If f be the greatest integer function and g be the modulus function, then

$(gof)\left(-\dfrac{5}{3}\right) - (fog)\left(-\dfrac{5}{3}\right) =$

(A) 1 (B) –1 (C) 2 (D) 4

Q29. Let $f(x) = \ln x$

& $g(x) = \dfrac{x^4 - x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 3}$.

The domain of f(g(x)) is -

(A) $(-\infty, \infty)$ (B) $[0, \infty)$

(C) $(0, \infty)$ (D) $[1, \infty)$

Q30. If a function f(x) is such that $f\left(x + \dfrac{1}{x}\right) = x^2 + \dfrac{1}{x^2}$; then $(fof)(\sqrt{11}) =$

(A) 9 (B) 81

(C) 79 (D) $\sqrt{11}$

Inverse of a Function

Q31. The values of the parameter α, for which the function $f(x) = 1 + \alpha x$, $\alpha \neq 0$ is the inverse of it self, is -

(A) 1 (B) 2 (C) –1 (D) 0

Q32. If $f(x) = 3x - 5$, then $f^{-1}(x)$ -

(A) is given by $\dfrac{1}{3x - 5}$

(B) is given by $\dfrac{x + 5}{3}$

(C) does not exist because f is not one–one

(D) does not exist because f is not onto

Q33. Which of the following functions is invertible -

(A) $f : R \to R$, $f(x) = e^x$

(B) $f : R \to R$, $f(x) = e^x + e^{-x}$

(C) $f : R \to R$, $f(x) = \sin^2 x + \cos^2 x$

(D) $f : R \to R$, $f(x) = 2x^3 - 1$

Odd & Even Functions

Q34. The function $f(x) = \log(x + \sqrt{x^2+1})$, is -

(A) neither an even nor an odd function

(B) an even function

(C) an odd function

(D) a periodic function

Periodic Function

Q35. If $f : R \to R$ is a function satisfying the property $f(x+1) + f(x+3) = K \; \forall \; x \in R$ then the period of $f(x)$ is -

(A) 4 (B) K

(C) 1 (D) π

Q36. Which of the following is not a periodic function-

(A) $\sin 2x + \cos x$ (B) $\cos\sqrt{x}$

(C) $\tan 4x$ (D) $\log\cos 2x$

Q37. The period of $\sin^2 x$ is-

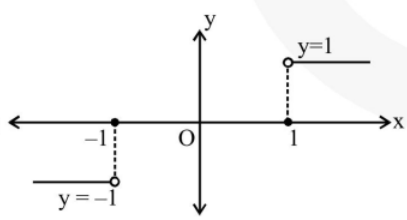
(A) $\pi/2$ (B) π

(C) $3\pi/2$ (D) 2π

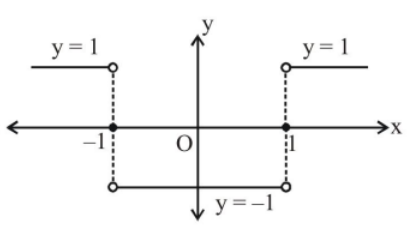
Transformation of Graph

Q38. Graph of **sgn** $(|x|-1)$ is -

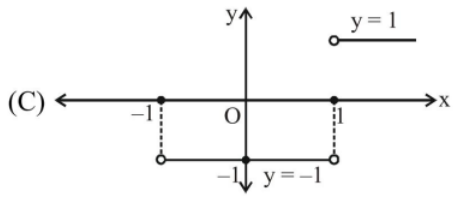
(A)



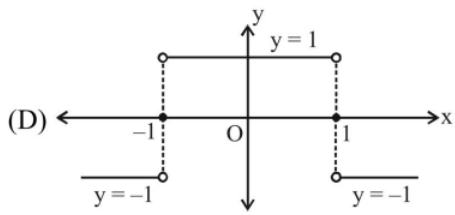
(B)



(C)



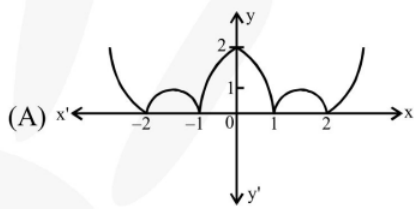
(D)



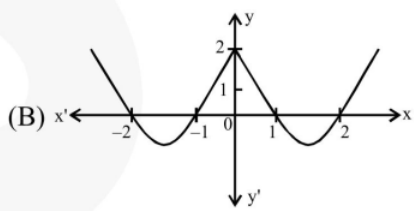
Q39. Graph of the function $f(x) = |x^2 - 3|x| + 2|$ is

($\because$ | | is modulus function)

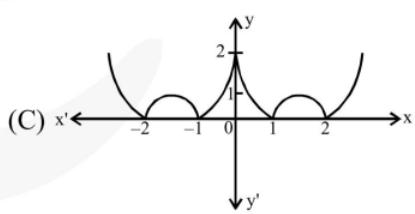
(A)



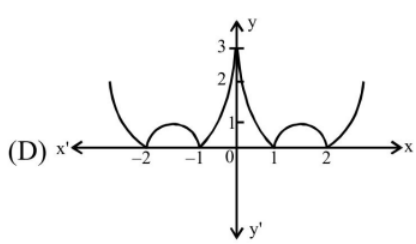
(B)



(C)



(D)



Exercise - 1A — Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If X = {a, b, c, d, e} and Y = {p, q, r, s, t} then which of the following subset(s) of X × Y is/are a function from X to Y

(A) {(a, r) (b, r) (b, s) (d, t) (e, q) (c, q)}

(B) {(a, r) (b, p) (c, t) (d, q)}

(C) {(a, p) (b, t) (c, r) (d, s) (e, q)}

(D) {(a, r) (b, r)(c, r) (d, r) (e, r)}

Q2. The values of x in $[-2\pi, 2\pi]$, for which the graph of the function $y = \sqrt{\frac{1+\sin x}{1-\sin x}} - \sec x$ and $y = -\sqrt{\frac{1-\sin x}{1+\sin x}} + \sec x$, coincide are

(A) $\left[-2\pi, -\frac{3\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right]$

(B) $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

(C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(D) $[-2\pi, 2\pi] - \left\{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}\right\}$

Q3. Which of the following function(s) have no domain?

(A) $f(x) = \log_{x-1}(2 - [x] - [x]^2)$ where [x] denotes the greatest integer function.

(B) $g(x) = \cos^{-1}(2-\{x\})$ where {x} denotes the fractional part function.

(C) $h(x) = \ln \ln(\cos x)$

(D) $f(x) = \frac{1}{\sec^{-1}\left(\text{sgn}\left(e^{-x}\right)\right)}$

Q4. The graphs of which of the following pairs of function differ.

(A) $y = \frac{\sin x}{\sqrt{1+\tan^2 x}} + \frac{\cos x}{\sqrt{1+\cot^2 x}}$;

$y = \sin 2x$

(B) $y = \tan x \cot x$;

$y = \sin x \operatorname{cosec} x$

(C) $y = |\cos x| + |\sin x|$;

$y = \frac{|\sec x| + |\operatorname{cosec} x|}{|\sec x| |\operatorname{cosec} x|}$

(D) none of these

Q5. If $g(x) = \left(4\cos^4 x - 2\cos 2x - \frac{1}{2}\cos 4x - x^7\right)^{\frac{1}{7}}$, then the value of $g(g(100))$ is equal to -

(A) 1000

(B) 10.g (g(10))

(C) 1

(D) 100

Q6. Which of the following function(s) would represent a non singular mapping.

(A) $f : R \to R \quad f(x) = |x| \text{ Sgn } x$ where Sgn denotes Signum function

(B) $g : R \to R \quad g(x) = x^{3/5}$

(C) $h : R \to R \quad h(x) = x^4 + 3x^2 + 1$

(D) $k : R \to R \quad k(x) = \frac{3x^2 - 7x + 6}{x - x^2 - 2}$

Q7. If $f(x) = x^2 + bx + c$ and $f(2 + t) = f(2 - t)$ for all real numbers t, then which of the following is INCORRECT.

(A) $f(1) < f(2) < f(4)$

(B) $f(2) < f(1) < f(4)$

(C) $f(2) < f(4) < f(1)$

(D) $f(4) < f(2) < f(1)$

Q8. Which of the following function(s) is/are periodic with period π ?

(A) $f(x) = |\sin x|$

(B) $f(x) = [x + \pi]$

(C) $f(x) = \cos(\sin x)$

(D) $f(x) = \cos^2 x$

(where [.] denotes the greatest integer function)

Q9. Consider the function g (x) defined as

$g(x)\cdot\left(x^{(2^{2008}-1)}-1\right) = (x+1)(x^2+1)(x^4+1).....$

$\left(x^{2^{2007}}+1\right)-1$. The value of g (2) is

(A) greater than g(4)

(B) greater than g(3)

(C) greater then g(-1)

(D) equal to 2

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 10 to 13)

Let $f(x) = x^2 - 2x - 1 \ \forall \ x \in R$.

Let $f: (-\infty, a] \to [b, \infty)$, where 'a' is the largest real number for which f (x) is bijective.

Q10. The value of (a + b) is equal to

(A) – 2

(B) – 1

(C) 0

(D) 1

Q11. Let $f: R \to R$, $g(x) = f(x) + 3x - 1$, then the least value of function $y = g(|x|)$ is

(A) – 9/4

(B) – 5/4

(C) – 2

(D) – 1

Q12. Let $f: [a, \infty) \to [b, \infty)$, then $f^{-1}(x)$ is given by

(A) $1 + \sqrt{x+2}$

(B) $1 - \sqrt{x+3}$

(C) $1 - \sqrt{x+2}$

(D) $1 + \sqrt{x+3}$

Q13. Let $f: R \to R$, then range of values of k for which equation $f(|x|) = k$ has 4 distinct real roots is

(A) (– 2, – 1)

(B) (– 2, 0)

(C) (– 1, 0)

(D) (0, 1)

Paragraph # 2 (Ques. 14 to 15)

Consider a quadratic function $f(x) = ax^2 + bx + c$, $(a, b, c \in R, a \neq 0)$ and satisfying the following conditions.

(i) $f(x - 4) = f(2 - x) \ \forall \ x \in R$ and $f(x) \geq x \ \forall \ x \in R$

(ii) $f(x) \leq \left(\frac{x+1}{2}\right)^2 \ \forall \ x \in (0, 2)$

(iii) The minimum value of f (x) is zero.

Q14. The value of the leading coefficient of the quadratic polynomial is -

(A) 1/4

(B) 1/3

(C) 1/2

(D) 1

Q15. f ' (1) has the value equal to -

(A) 1/4

(B) 1/3

(C) 1/2

(D) 1

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R),(S) and (T). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

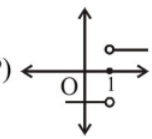
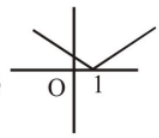
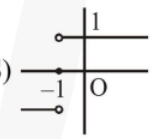
Q16. Match the entries of column I with one or more than one entries of column II. Note that [x], {x} and sgn(x) denotes the largest integer less than or equal to x, fractional part of x and signum function of x respectively.

Column-I	**Column-II**
(A) Let $f:[-1,1]\to R$ be defined by $f(x)=\sqrt[5]{x}+\sin^{-1}x$ then f(x) is	(P) Odd
(B) Let $f:R\to\{-1,0,1\}$ is defined by $f(x)=\text{sgn}\left(\frac{1-\lvert x\rvert}{1+\lvert x\rvert}\right)$ then f(x) is	(Q) Even
(C) Let $f:(-\infty,0]\to[0,\infty)$ be defined by $f(x)=\frac{2^{-[x]}}{2^{\{x\}}}-2^{\lvert x\rvert}$ then f(x) is	(R) Onto
(D) Let $f:(-4,2]\to[0,3]$ be defined by $f(x)=\sqrt{8-2x-x^2}$ then f(x) is	(S) One-one
	(T) Many-one

Q17.

Column-I	**Column-II**
(A) Domain of $\sin^{-1}(5x)$	(P) 1
(B) Domain of $\sqrt{1-25x^2}$	(Q) $\frac{x+1}{x-1}$
(C) If $f(x)=\frac{x+1}{x-1}, x\neq 1$ then (fofof) (x) =	(R) $\left[-\frac{1}{5},\frac{1}{5}\right]$
(D) Period of $x-[x]$ is	(S) $\left[0,\frac{1}{5}\right]$
	(T) x

Q18. In this question C-I contains function and C-II contains graphs, match correct function in C-I with correct graph in C-II

Column-I	**Column-II**
(A) $y = \lvert x + 1 \rvert$	(P) 
(B) $y = \lvert x - 1 \rvert$	(Q) 
(C) $y = \text{sgn}(x + 1)$	(R) 
(D) $y = \text{sgn}(x - 1)$	(S) 

Q19.

Column-I	**Column-II**
(A) Let $f : [-1, \infty) \to (0, \infty)$ defined by $f(x) = e^{x^2+\lvert x \rvert}$ then $f(x)$, is	(P) one-one
(B) Let $f : (1, \infty) \to [3, \infty)$ defined by $f(x) = \sqrt{10 - 2x + x^2}$, then $f(x)$ is	(Q) into
(C) Let $f : R \to I$ defined by $f(x) = \tan^5\pi[x^2 + 2x + 3]$ where [ ] denotes greatest integer function, then $f(x)$ is	(R) many one
(D) Let $f : [3, 4] \to [4, 6]$ defined by $f(x) = \lvert x-1 \rvert + \lvert x-2 \rvert + \lvert x-3 \rvert + \lvert x-4 \rvert$ then $f(x)$	(S) onto (T) periodic

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q20. Suppose f is a real function satisfying $f(x + f(x)) = 4f(x)$ and $f(1) = 4$.
Find the value of $f(21)$.

Q21. Let f be function defined from $R^+ \to R^+$. If $[f(xy)]^2 = x(f(y))^2$ for all positive numbers x and y and $f(2) = 6$, find the value of $f(50)$.

Q22. Let f(x) be a function with two properties
(i) for any two real number x and y, $f(x + y) = x + f(y)$ and
(ii) $f(0) = 2$.
Find the value of $f(100)$.

SUBJECTIVE TYPE

Q23. Find domain of $2^x + 2^y = 2$

Q24. Find domain of function defined by

$$f(x) = \sqrt{\log\left(\frac{5x - x^2}{6}\right)}.$$

Q25. Find the domains of definitions of the following functions.

(Read the symbols [*] and {*} as greatest integers and fractional part functions respectively.)

(i) $f(x) = \sqrt{\cos 2x} + \sqrt{16 - x^2}$

(ii) $f(x) = \ln\left(\sqrt{x^2 - 5x - 24} - x - 2\right)$

(iii) $f(x) = \sqrt{\frac{1 - 5^x}{7^{-x} - 7}}$

(iv) $f(x) = \frac{1}{\sqrt{4x^2 - 1}} + \ln x(x^2 - 1)$

(v) $f(x) = \sqrt{x^2 - |x|} + \frac{1}{\sqrt{9 - x^2}}$

Q26. If $y = 2[x] + 3$ & $y = 3[x - 2] + 5$, then find $[x + y]$

where [.] denotes greatest integer

Q27. If $0 \le x \le 1.5$ then range of $[x^2]$

Q28. If $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$ then range of $\sin[x]$

Q29. Find range of $(-1)^{[x]}$, (where [.] denotes greatest integer function-

Q30. Find the values of

$$\left[\frac{3}{4}\right] + \left[\frac{3}{4} + \frac{1}{100}\right] + \left[\frac{3}{4} + \frac{2}{100}\right] + \ldots\ldots \left[\frac{3}{4} + \frac{99}{100}\right]$$

([.] denotes g.I.f.)

Q31. Find the range of the function: $y = 9 + x^2$

Q32. Find the range of $f(x) = 3 + x - [x + 2]$

Q33. If $A = \{0, 2\}$, $B = \{0, 4\}$, then find whether $f : A \to B$, $f(x) = 2x$ and $g : A \to B$, $g(x) = x^2$ are equal functions or not.

Q34. Show $f : R \to R$ defined by $f(x) = x^2 + x$ for all $x \in R$ is many-one function.

Q35. If $A = \{2, 3, 4\}$ and $B = \{3, 4, 5, 6, 7\}$ and a function from A to B is defined by the rule $f(x) = 2x - 1$ then find whether a given function is one-one or not ?

Q36. Find the number of functions

$f : \{a, b, c\} \to \{1, 2, 3, 4, 5, 6\}$ which are not injective.

Q37. Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ 1 & x > 0 \end{cases}$

then for all x find f(g(x))

Q38. Let $f(x) = \begin{cases} 1 + x, & 0 \le x \le 2 \\ 3 - x, & 2 < x \le 3 \end{cases}$, then find (fof)(x).

Q39. If $f : R - [-3/2, \infty) \to [19/4, \infty)$ where $f(x) = x^2 + 3x + 7$ then find $f^{-1}(7)$.

Q40. If $f : R \to R$, $f(x) = x^3 + 2$ then find $f^{-1}(x)$

Q41. The function $f(x) = \frac{3x - 2}{x + 4}$ has an inverse that can be written in the form

$$f^{-1}(x) = \frac{x + b}{cx + d}.$$

Find the values of b, c, d.

Q42. If $f(x) = \frac{a^x + 1}{a^x - 1}$ then find whether it is an even or odd function.

Q43. Suppose that f (x) is a function of the form

$$f(x) = \frac{ax^8 + bx^6 + cx^4 + dx^2 + 15x + 1}{x} (x \ne 0).$$

If $f(5) = 2$ then find the value of $f(-5)$.

Q44. Suppose that f is an even, periodic function with period 2, and that f (x) = x for all x in the interval [0, 1]. Find the value of f (3.14).

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If $f(x) = \log\left(\frac{1+x}{1-x}\right)$ when $-1 < x_1, x_2 < 1$, then $f(x_1) + f(x_2)$ equals

(A) $f\left(\frac{x_1+x_2}{1+x_1x_2}\right)$ (B) $f\left(\frac{x_1-x_2}{1+x_1x_2}\right)$

(C) $f\left(\frac{x_1+x_2}{1-x_1x_2}\right)$ (D) $f\left(\frac{x_1-x_2}{1-x_1x_2}\right)$

Q2. The value of b and c for which the identity $f(x+1) - f(x) = 8x + 3$ is satisfied, where $f(x) = bx^2 + cx + d$, are –

(A) $b = 2, c = 1$ (B) $b = 4, c = -1$

(C) $b = -1, c = 4$ (D) $b = -1, c = 1$

Q3. The domain set of definition of $f(x) = \sqrt{\frac{1-|x|}{2-|x|}}$ is -

(A) $(-\infty, \infty) - [-2, 2]$

(B) $(-\infty, \infty) - [-1, 1]$

(C) $[-1,1] \cup (-\infty, -2) \cup (2, \infty)$

(D) none of these

Q4. If $b^2 - 4ac = 0$, $a > 0$, then the domain of the function $f(x) = \log(ax^3 + (a+b)x^2 + (b+c)x + c)$ is-

(A) $R - \left\{-\frac{b}{2a}\right\}$

(B) $R - \left\{\left\{-\frac{b}{2a}\right\} \cup \{x \mid x \geq -1\}\right\}$

(C) $R - \left\{-\frac{b}{2a}\right\} \cap (-1, \infty)$

(D) none of these.

Q5. The function $f(x) = \sqrt{\cos(\sin x)} + \sin^{-1}\left(\frac{1+x^2}{2x}\right)$ is defined for

(A) $x \in \{-1, 1\}$ (B) $x \in [-1, 1]$

(C) $x \in R$ (D) $x \in (-1, 1)$

Q6. Domain of the function $y = \sqrt{\log_{10}(\log_{10} x) - \log_{10}(4 - \log_{10} x) - \log_{10} 3}$ is -

(A) $[0, 10^4)$ (B) $(10^3, 10^4)$

(C) $(0, 10^4)$ (D) $[10^3, 10^4)$

Q7. The largest interval lying in $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ for which the function $\left[f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x)\right]$ is defined, is

(A) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (B) $\left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$

(C) $\left[0, \frac{\pi}{2}\right)$ (D) $[0, \pi]$

Q8. The range of the function $y = \frac{x^2}{1+x^2}$ is -

(A) [0, 1) (B) [0, 1]

(C) (0, 1] (D) None of these

Q9. The range of the function $f(x) = \log_e(3x^2 - 4x + 5)$ is -

(A) $\left(-\infty, \log_e \frac{11}{3}\right]$

(B) $\left[\log_e \frac{11}{3}, \infty\right)$

(C) $\left[-\log_e \frac{11}{3}, \log_e \frac{11}{3}\right]$

(D) None of these

Q10. The range of the function $f(x)=\frac{x^2-x}{x^2+2x}$ is-

(A) $R\in\left\{1,-\frac{1}{2}\right\}$ (B) R

(C) R\{1} (D) none of these

Q11. The range of the function $y=\frac{1}{2-\sin 3x}$ is -

(A) $\left(\frac{1}{3},1\right)$ (B) $\left[\frac{1}{3},1\right)$

(C) $\left[\frac{1}{3},1\right]$ (D) None of these

Q12. Let f be a real valued function satisfying $f(x+y)=f(x).f(y)$ for all $x, y \in R$ and $f(1)=2$.

If $\sum_{k=1}^{n} f(a+k) = 16(2^n-1)$, then a = ?

(A) 3 (B) 4

(C) 2 (D) None

Q13. If $af(x+1)+bf\left(\frac{1}{x+1}\right)=x$, $x\neq -1$, $a\neq b$, then f(1) is equal to

(A) a + b (B) a^2-b^2

(C) $\left(\frac{1}{a+b}\right)$ (D) 0

Q14. A real valued function $f(x)$ satisfies the functional equation

$f(x-y)=f(x)f(y)-f(a-x)\,f(a+y)$

where a is a given constant and $f(0)=1$, $f(2a-x)$ is equal to -

(A) $f(1)+f(a-x)$ (B) $f(-x)$

(C) $-f(x)$ (D) $f(x)$

Q15. The function f satisfies the functional equation $3f(x)+2f\left(\frac{x+59}{x-1}\right)=10x+30$ for all real $x\neq 1$. The value of f(7) is

(A) 8 (B) 4 (C) –8 (D) 11

Q16. If $f: R\to R$ and $f(x+y)=x\cdot f(y)+y\cdot f(x)$ then $f(21)$ is -

(A) 0 (B) 44

(C) 21 (D) 63

Q17. If $f: R\to S$ defined by

$f(x)=\sin x-\sqrt{3}\cos x+1$

is onto, then the interval of S is-

(A) [–1, 3] (B) [–1, 1]

(C) [0, 1] (D) [0, –1]

Q18. If functions f (x) and g (x) are defined on $R\to R$ such that

$$f(x)=\begin{cases}0, & x\in \text{rational}\\ x, & x\in \text{irrational}\end{cases},$$

$$g(x)=\begin{cases}0, & x\in \text{irrational}\\ x, & x\in \text{rational}\end{cases},$$

then (f – g) (x) is -

(A) one-one and onto

(B) neither one-one nor onto

(C) one-one but not onto

(D) onto but not one-one

Q19. If $f: R-\{3\}\longrightarrow R-\{1\}$; $f(x)=\left(\frac{x-2}{x-3}\right)$ is -

(A) one-one & into

(B) one-one & onto

(C) many-one & into

(D) many-one & onto

Q 20. Let $f: X\to Y$ be a function such that $f(x)=\sqrt{x-2}+\sqrt{4-x}$, then the set of X and Y for which f(x) is both injective as well as surjective, is-

(A) [2,4] and $\left[\sqrt{2},2\right]$

(B) [3,4] and $\left[\sqrt{2},2\right]$

(C) [2,4] and [1,2]

(D) [2,3] and [1,2]

Q21. If $f(x, y) = (\max(x,y))^{\min(x,y)}$ and $g(x, y) = \max(x, y) - \min(x, y)$, then $f\left(g\left(-1, -\frac{3}{2}\right), g(-4, -1.75)\right)$ equals-

(A) –0.5 (B) 0.5
(C) 1 (D) 1.5

Q22. If $f : [1, \infty) \to [2, \infty)$ is given by, $f(x) = x + \frac{1}{x}$, then $f^{-1}(x)$ equals :

(A) $\frac{x + \sqrt{x^2 - 4}}{2}$ (B) $\frac{x}{1 + x^2}$

(C) $\frac{x - \sqrt{x^2 - 4}}{2}$ (D) $1 - \sqrt{x^2 - 4}$

Q23. Function $f(x) = \log_e(x^3 + \sqrt{1 + x^6})$ is -

(A) even
(B) odd
(C) neither even nor odd
(D) none of these

Q24. Which of the following function has a period of 2π?

(A) $f(x) = \sin\left(2\pi x + \frac{\pi}{3}\right) + 2\sin\left(3\pi x + \frac{\pi}{4}\right) + 3\sin 5\pi x$

(B) $f(x) = \sin\frac{\pi x}{3} + \sin\frac{\pi x}{4}$

(C) $f(x) = \sin x + \cos 2x$

(D) none

Q25. Period of $f(x) = nx + n - [nx + n]$, $(n \in N$ where $[\cdot]$ denotes the greatest integer function is -

(A) 1 (B) $\frac{1}{n}$
(C) n (D) None of these

Q26. The period of the function

$f(x) = \cos\left(\frac{\pi x}{n!}\right) - \sin\left(\frac{\pi x}{(n+1)!}\right)$ is

(A) 2(n + 1)! (B) 2(n!)
(C) (n + 1) (D) not periodic

Q27. If f(x) is an odd periodic function with period 2, then f(4) equals

(A) –4 (B) 4
(C) 2 (D) 0

28. If $x^4 f(x) - \sqrt{1 - \sin 2\pi x} = |f(x)| - 2f(x)$, then f(– 2) equals :-

(A) $\frac{1}{17}$ (B) $\frac{1}{11}$

(C) $\frac{1}{19}$ (D) 0

Q29. Let $f(x) = (2\cos x - 1)(2\cos 2x - 1)(2\cos 4x - 1)$ ..... $(2\cos 32x - 1)$, then $f\left(\frac{2\pi}{13}\right)$ is -

(A) –1 (B) 1
(C) $\frac{1}{2}$ (D) None

Exercise - 2A

Brain Booster

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following function are defined for all x

(A) sin [x] + cos [x] ([x] denotes greatest integer $\le$ x)

(B) $\sec^{-1}\left(1+\sin^2 x\right)$

(C) tan(log x)

(D) $\sqrt{\frac{9}{8}+\cos x+\cos 2x}$

Q2. Which of the following pairs of functions are identical ?

(A) $f(x) = \log_x e$, $g(x) = \frac{1}{\log_e x}$

(B) $\text{sgn}(x^2 + 1)$; $g(x) = \sin^2 x + \cos^2 x$

(C) $f(x) = \sec^2 x - \tan^2 x$; $g(x) = \text{cosec}^2 x - \cot^2 x$

(D) $f(x) = \frac{1}{|x|}$; $g(x) = \sqrt{x^{-2}}$

Q3. Which of the following functions are not homogeneous ?

(A) $x + y\cos\frac{y}{x}$

(B) $\frac{xy}{x+y^2}$

(C) $\frac{x - y\cos x}{y\sin x + y}$

(D) $\frac{x}{y}\ell n\left(\frac{y}{x}\right)+\frac{y}{x}\ell n\left(\frac{x}{y}\right)$

Q4. Let $f : R \to R$ and $g : R \to R$ be defined by $f(x) = [x]$ and $g(x) = \frac{3-2x}{4}$ then

(A) f is neither one-one nor onto

(B) g is one-one but f is not one-one

(C) f is one-one and g is onto

(D) neither f nor g is onto.

Q5. Which of the functions defined below are NOT one-one function(s) ?

(A) $f(x) = 5(x^2 + 4)$, $(x \in R)$

(B) $g(x) = 2x+(1/x)$

(C) $h(x) = \ell n(x^2+x+1)$, $(x \in R)$

(D) $f(x) = e^{-x}$

Q6. Let $f(x)$ be invertible function and let $f^{-1}(x)$ be its inverse.

Let equation $f\left\{f^{-1}(x)\right\} = f^{-1}(x)$ has two real roots α and β (within domain of $f(x)$). Then -

(A) $f(x) = x$ also have same two real roots.

(B) $f^{-1}(x) = x$ also have same two real roots

(C) $f(x) = f^{-1}(x)$ also have same two real roots

(D) Area formed by $(0,0)$ $(\alpha, f(\alpha))$ and $(\beta, f(\beta))$ is 1 unit.

Q7. Which of the functions are even -

(A) $\log\left(\frac{1+x^2}{1-x^2}\right)$

(B) $\sin^2 x + \cos^2 x$

(C) $\log\left(\frac{1+x^3}{1-x^3}\right)$

(D) $\frac{(1+2^x)^2}{2^x}$

Q8. Let f (x) be a real valued ,function defined as $f(x) = \left[e^{|\sin p\pi x|}\right]$ where p is a +ve integer and [x] represents greatest integer less than or equal to x, then -

(A) Range of f (x) is {1, 2}

(B) Period of f(x) is $\frac{1}{p}$

(C) Domain of f (x) is R

(D) f (x) is not periodic.

Q9. $f(x) = \cos^2 x + \cos^2\left(\frac{\pi}{3}+x\right) - \cos x \cos\left(\frac{\pi}{3}+x\right)$ is -

(A) an odd function

(B) an even function

(C) a periodic function

(D) f (0) = f (1)

Q10. The equation $\sin x - \frac{\pi}{2} + 1 = 0$ has one root in interval.

(A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \pi\right)$

(C) $\left(\pi, \frac{3\pi}{2}\right)$ (D) None of these

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are Three questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 11 to 13)

Least positive integral solution (x_1) of the inequality $\frac{5x+8}{4-x} < 2$ is a root of the equation f(x) = 0, where $f(x) = \frac{a}{5}x^2 - \left(\frac{a^2-1}{5}\right)x - 1, a \in [0,1]$. Then

Q11. Possible value of x_1 is

(A) 0 (B) 4

(C) 5 (D) 1

Q12. Possible value of 'a' is/are

(A) 0 (B) 0, 5

(C) $\frac{1}{2}$ (D) none of these

Q13. The inverse of f(x) is

(A) $\frac{x}{5} - 1$ (B) 5x + 1

(C) 5x – 5 (D) 5x + 5

Paragraph # 2 (Ques. 14 to 16)

Consider the function $f(x) = \begin{cases} x - [x] - \frac{1}{2}; & \text{if } x \notin I \\ 0 \quad ; & \text{if } x \in I \end{cases}$

where [.] denotes greatest integer function.

If g(x) = max. $\{x^2, f(x), |x|\}$; $-2 \le x \le 2$, then.

Q14. Range of f(x) is-

(A) [0, 1) (B) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(C) $\left(-\frac{1}{2}, \frac{1}{2}\right)$ (D) $\left[-\frac{1}{2}, \frac{1}{2}\right)$

Q15. f(x) is-

(A) non periodic

(B) periodic with period 1

(C) periodic with period 2

(D) periodic with period 1/2

Q16. The set of values of a, if g(x) = a has three real and distinct solutions, is -

(A) (0, 1/2) (B) (0, 1/4)

(C) (1/4, 1/2) (D) (0, 1)

Paragraph # 3 (Ques. 17 to 18)

Let, $f(x) = (\cos x) + \frac{1}{2}, \quad x \in [0, 2\pi]$

$g(x) = (\sin x) + \frac{3}{5}, \quad x \in [0, \pi]$

Q17. Number of integral solutions of the equation $[f(x)] = 0$ is -

[Note : $[\cdot]$ denotes greatest integer function]

(A) 0 (B) 2 (C) 3 (D) 4

Q18. Number of solutions of the equation $\{g(x)\} = \frac{1}{2}$ is -

[Note : $\{\cdot\}$ denotes fractional part function]

(A) 0 (B) 2

(C) 3 (D) 4

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q19.

Column-I $f(x)$	**Column-II** Range
(A) $\frac{\cos^2 x + \cos x + 2}{\cos^2 x + \cos x + 1}$	(P) $\left(0, \frac{7}{3}\right]$
(B) $\left\lvert\frac{(\sqrt{\cos x} - \sqrt{\sin x})(\sqrt{\cos x} + \sqrt{\sin x})}{3(\cos x + \sin x)}\right\rvert$	(Q) $\left[\frac{4}{3}, \frac{7}{3}\right]$
(C) $\frac{7}{3(x^6 + 2x^4 + 3x^2 + 1)}$	(R) $\left[0, \frac{1}{3}\right]$
(D) $\log_8(x^2 + 2x + 2)$	(S) $[0, \infty)$

Q20.

Column-I	**Column-II**
(A) $f : R \to R$, $f(x) = (x - 1)(x - 2).....(x - 11)$	(P) one one
(B) $f : R - \{-4/3\} \to R$, $f(x) = \frac{2x + 1}{3x + 4}$	(Q) onto
(C) $f : R \to R$, $f(x) = e^{\sin x} + e^{-\sin x}$	(R) into
(D) $f : R \to R$, $f(x) = \log(x^2 + 2x + 3)$	(S) into

Q21.

Column-I	Column-II
(A) Let continuous $f : R \to R$ satisfies $f(x + y) + f(x - y) = 2f(x)\, f(y)\ \forall$ $x, y \in R$ and $f(0) \neq 0$, then $f(x)$ is	(P) Even
(B) Let $f : R \to R$ is defined by $f(x) = \frac{e^{\lvert x\rvert} - e^{-x}}{e^{x} + e^{-x}}$, then $f(x)$ is	(Q) Odd
(C) Let $f : R \to R$ be a polynomial function satisfying $f(x)\ f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ and $f(3) = 28$ then $f(x)$ is	(R) If(x)
	(S) Manyone
(D) Let $f : R \to R$ is defined by $f(x) = 2x + \sin x$, then $f(x)$ is	(T) Bijective

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q22. Let $f(x) = (x + 1)(x + 2)(x + 3)(x + 4) + 5$ where $x \in [-6, 6]$. If the range of the function is $[a, b]$ where $a, b \in N$ then find the value of $(a + b)$.

Q23. Number of values of x, for which $\{x\}$, sgn(x) and $[x]$ are in an A.P. is ?
(where $\{.\}$ is fractional part function, $[.]$ is greatest integer function and sgn(x) is signum function)

Q24. Let $f(x) = -x^{100}$. If $f(x)$ is divided by $x^2 + x$, then the remainder is $r(x)$. Find the value of $r(10)$.

Q25. Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$, where $a, b, c, d \in R$.
Suppose $P(0) = 6$, $P(1) = 7$, $P(2) = 8$ and $P(3) = 9$, then find the value of $P(4)$.

Q26. The polynomial $R(x)$ is the remainder upon dividing x^{2007} by $x^2 - 5x + 6$. If $R(0)$ can be expressed as $ab(a^c - b^c)$, find the value of $(a + b + c)$.

Q27. If $f : R \to R$, $f(x) = x^3 + x^2 + 3x + 5$, then the value of $f^{-1}(50)$ is ?

Q28. $f(x)$ and $g(x)$ are linear function such that for all x, $f(g(x))$ and $g(f(x))$ are Identity functions. If $f(0) = 4$ and $g(5) = 17$, compute $f(136)$.

Q29. Let a and b be real numbers and let $f(x) = a \sin x + b \sqrt[3]{x} + 4$, $\forall\ x \in R$.
If $f\left(\log_{10}(\log_3 10)\right) = 5$ then find the value of $f\left(\log_{10}(\log_{10} 3)\right)$.

Q30. Let $f(k) = (k^2 + 3k + 2)^{\cos(\pi k)}$, then the value of $\sum_{k=1}^{18} \log_{10} f(k)$ is ?

Q31. The number of even integral value(s) in the range of the function
$f(x) = \frac{\tan^2 x + 8\tan x + 15}{1 + \tan^2 x}$ is ?

SUBJECTIVE TYPE

Q32. Find the domain $f(x) = \frac{1}{\sqrt{[[|x|-5]]-11}}$ where [.] denotes greatest integer function.

Q33. Find the domains of definitions of the following functions.
(Read the symbols [*] and {*} as greatest integers and fractional part functions respectively.)
(i) $f(x) = \log_7 \log_5 \log_3 \log_2 (2x^3 + 5x^2 - 14x)$
(ii) $y = \log_{10} \sin(x-3) + \sqrt{16 - x^2}$
(iii) $f(x) = \sqrt{(x^2 - 3x - 10).\ln^2(x-3)}$

Q34. If $f(x + y) = f(x) + f(y)$, $f(1) = 7$ then find summation of $\sum_{n=1}^{n} f(r)$.

Q35. If $f(x + y, x - y) = x.y$ then find A.M. of $f(x, y)$ and $f(y, x)$.

Q36. If $f(x) = \log(x + \sqrt{1+x^2})$ then find $f^{-1}(x)$, where $f: R \to R^+$

Q37. The period of
$$f(x) = \frac{\sin 8x \cos x - \sin 6x.\cos 3x}{\cos 2x.\cos x - \sin 3x.\sin 4x}.$$

Q38. Draw the graph of $y = |2 - |x - 1||$.

Q39. Draw the graph of $y = |e^{|x|} - 2|$.

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
A	D	C	C	B	B	B	B	C	A
11	12	13	14	15	16	17	18	19	20
B	C	C	C	D	C	C	D	B	C
21	22	23	24	25	26	27	28	29	30
C	C	D	C	A	D	D	A	A	C
31	32	33	34	35	36	37	38	39	
C	B	D	C	A	B	B	B	C	

Ex - 1A

1. C,D **2.** A,C **3.** A,B,C,D **4.** A,B,C **5.** B,D
6. A,B **7.** A,C,D **8.** A,C,D **9.** A,B,C,D **10.** B
11. C **12.** A **13.** A **14.** A **15.** D

16. (A) → (P,S), (B) → (Q,R,T), (C) → (T), (D) → (R,T)

17. (A) → (R), (B) → (R), (C) → (Q), (D) → (P)

18. (A) → (R), (B) → (Q), (C) → (S), (D) → (P)

19. (A) → (Q,R), (B) → (P,Q), (C) → (Q,R,T), (D) → (P,S)

20. 64 **21.** 30 **22.** 102

23. Domain = $x \in (-\infty, 1)$ **24.** $x \in [2, 3]$

25. **(i)** $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right] \cup \left[-\frac{5\pi}{4}, -\frac{3\pi}{4}\right]$ **(ii)** $x \in (-\infty, -3]$

(iii) $x \in (-\infty, -1) \cup [0, \infty)$ **(iv)** $x \in \left(-1, -\frac{1}{2}\right) \cup (1, \infty)$

(v) $x \in (-3, -1] \cup [1, 3) \cup \{0\}$

26. 15 **27.** {0, 1, 2} **28.** [sin(–2), sin(–1) sin(0), sin(1)}

29. {–1, 1} **30.** 75 **31.** $[9, \infty)$ **32.** [1, 2)

33. f and g are equal functions **35.** Function is one-one **36.** 96

37. f(g(x)) = 1 for all $x \in R$ **38.** $\begin{cases} 2+x, & 0 \le x < 1 \\ 2-x, & 1 < x \le 2 \\ 4-x, & 2 < x \le 3 \end{cases}$ **39.** {–3}

40. $(x-2)^{1/3}$ **41.** $b = \frac{1}{2}, c = \frac{-1}{4}, d = \frac{3}{4}$ **42.** f(x) is an odd function

43. 28 **44.** 0.86

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	B	C	C	A	D	C	A	B	A
11	12	13	14	15	16	17	18	19	20
C	A	D	C	B	A	A	A	B	A
21	22	23	24	25	26	27	28	29	
D	A	B	C	B	A	D	A	B	

Ex - 2A

1. A,B,D **2.** A,B,D **3.** B,C **4.** A,B **5.** A,B,C

6. A,B **7.** A,B,D **8.** A,B,C **9.** B,C,D **10.** A,B

11. C **12.** A **13.** D **14.** C **15.** B

16. C **17.** B **18.** B

19. (A) → (Q), (B) → (R), (C) → (P), (D) → (S)

20. (A) → (Q,R), (B) → (P,S), (C) → (R,S), (D) → (R,S)

21. (A) → (P,R,S), (B) → (R,S), (C) → (T), (D) → (Q,T)

22. 5049 **23.** 3 **24.** 10 **25.** 34 **26.** 2011

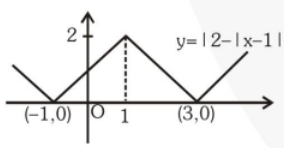
27. 3 **28.** 12 **29.** 3 **30.** 1

31. 9 **32.** $x \in (-\infty, -17] \cup [17, \infty)$

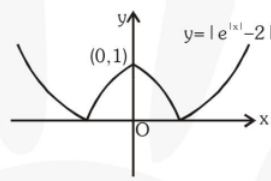
33. **(i)** $x \in \left(-4, -\frac{1}{2}\right) \cup (2, \infty)$ **(ii)** $(3-2\pi, 3-\pi) \cup (3, 4]$ **(iii)** $[5, \infty) \cup \{4\}$

34. $\frac{7n(n+1)}{2}$ **35.** 0 **36.** $\frac{e^x - e^{-x}}{2}$ **37.** Period $\frac{\pi}{2}$

38.



39.



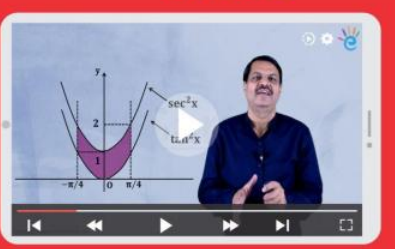
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